

Motor Trend: Transmission & MPG

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MOTOR TREND analysis

Data & Methods

- Dataset: mtcars
- Outcome: mpg
- Potential confounders: wt, hp , cyl, disp, gear, carb

Exploratory Data Analysis (EDA)

- **Descriptive statistics**

```
# summary stats of group per am
```

```
##      am          n mean_mpg sd_mpg mean_wt sd_wt mean_hp sd_hp mean_cyl
sd_cyl
## 1 Automatic    19      17.1   3.83    3.77 0.777    160.   53.9     6.95
1.54
## 2 Manual       13      24.4   6.17    2.41 0.617    127.   84.1     5.08
1.55
```

- **Boxplot MPG vs transmission type**

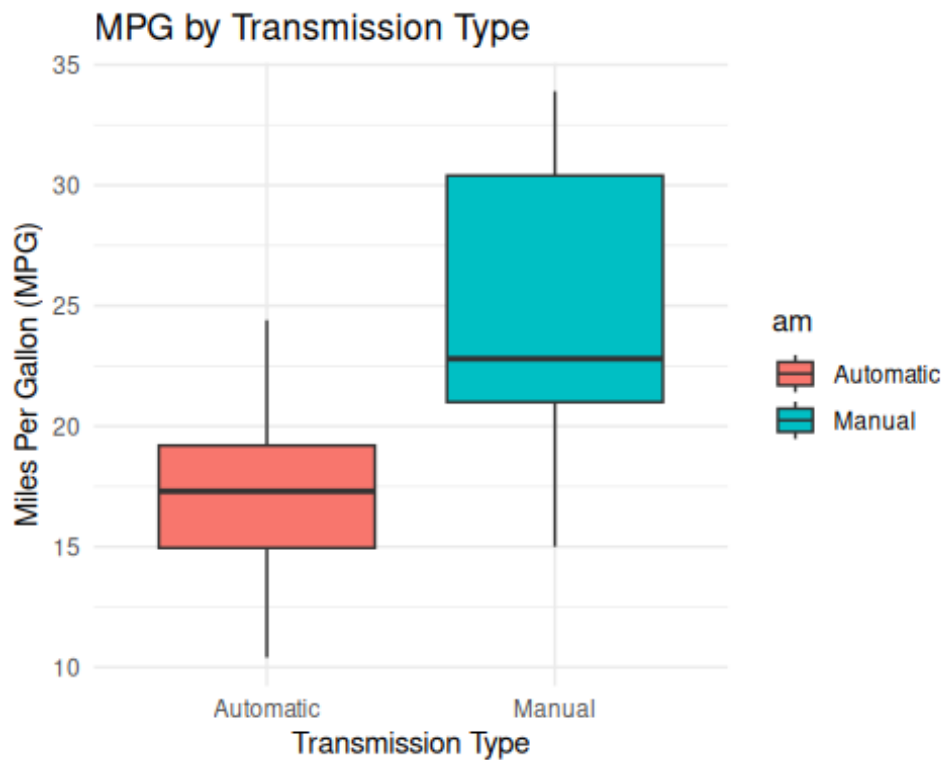


Figure 1 shows a clear separation between groups: manual cars tend to have higher MPG values and more variability compared with automatic cars.

- **Scatter Plot MPG vs wt**

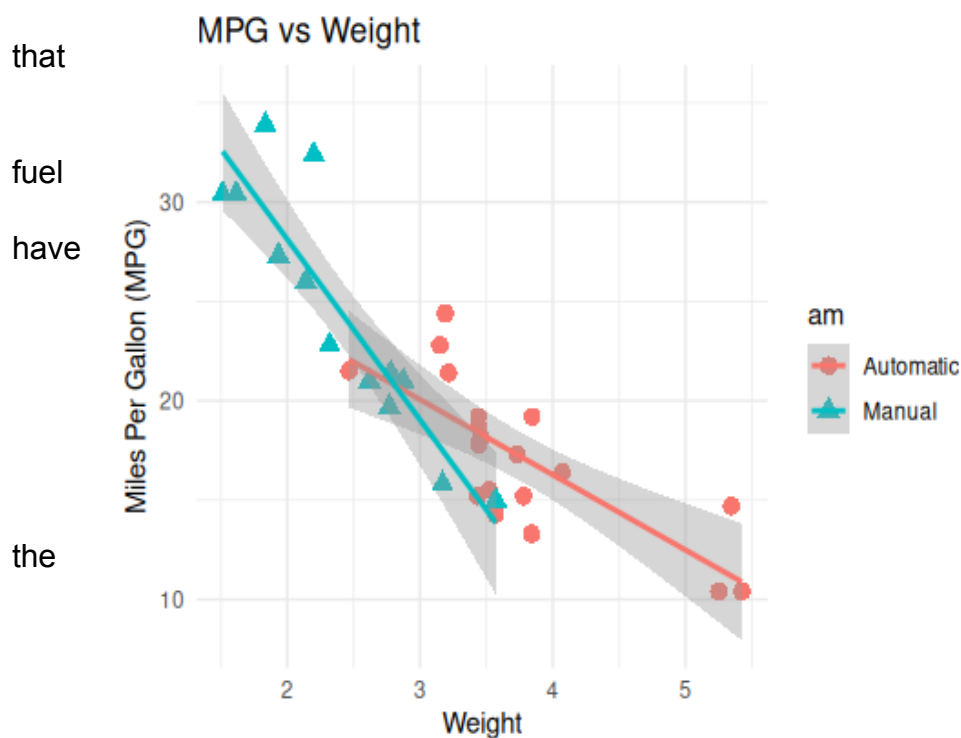


Figure 2 suggests the weight is strongly associated with efficiency: heavier cars a lower MPG. Manual cars are generally lighter, which may partly explain their higher fuel efficiency. This could suggest need for adjusted models

controlling weight as a confounder

Models

- **Model 1**

```
l## # A tibble: 2 × 7
##   term          estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)    17.1      1.12     15.2  1.13e-15    14.9     19.4
## 2 amManual       7.24      1.76      4.11  2.85e- 4     3.64     10.8
```

amManual coefficient reveal that there is a mpg mean difference between Manual and Automatic with a p.value < 0.05

- **Model 2 adjusted**

```
## # A tibble: 3 × 7
##   term          estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)   37.3      3.05     12.2  5.84e-13    31.1     43.6
## 2 amManual     -0.0236    1.55    -0.0153 9.88e- 1    -3.18     3.14
## 3 wt          -5.35      0.788    -6.79  1.87e- 7    -6.96    -3.74
```

We can see the Simpson Paradox in action: the amManual coefficient changed from 7.24 to -1.12 when adjusting for wt, meaning that wt is a confounder

- **Model 3 adjusted.**

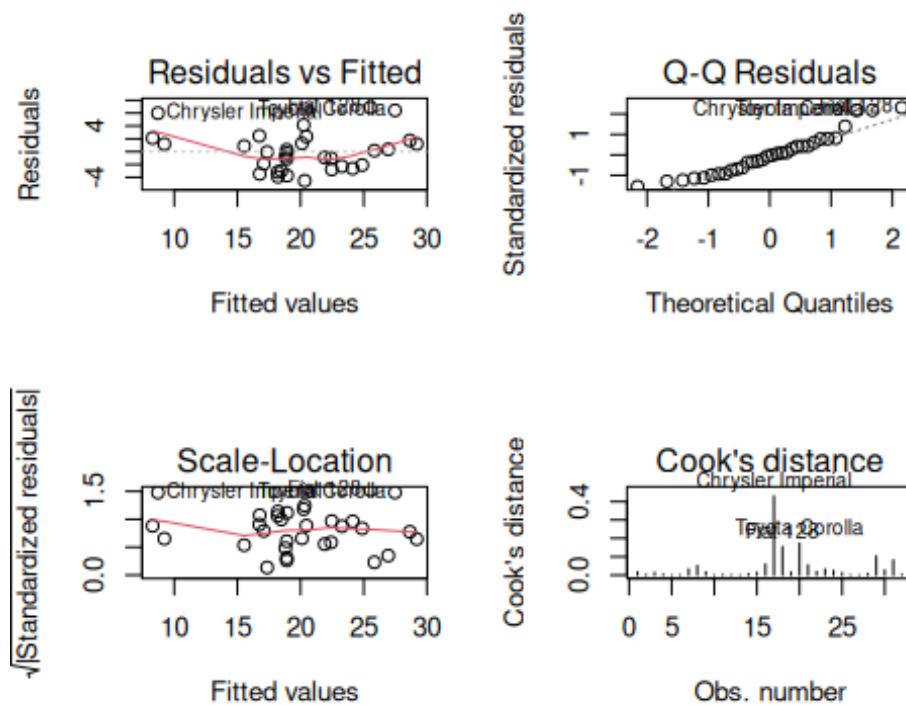
```
## # A tibble: 4 × 7
##   term          estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)   34.0      2.64     12.9  2.82e-13    28.6     39.4
## 2 amManual       2.08      1.38      1.51  1.41e- 1    -0.736     4.90
## 3 wt            -2.88      0.905    -3.18  3.57e- 3    -4.73     -1.02
## 4 hp            -0.0375    0.00961   -3.90  5.46e- 4    -0.0572    -0.0178
```

Comparing the model to evaluate the confounding effect of wt and hp on the am coefficient. ANOVA between the 3 models

```
## Analysis of Variance Table
##
## Model 1: mpg ~ am
## Model 2: mpg ~ am + wt
## Model 3: mpg ~ am + wt + hp
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      30 720.90
## 2      29 278.32  1    442.58 68.734 5.071e-09 ***
## 3      28 180.29  1     98.03 15.224 0.0005464 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p.value of the ANOVA test between model 1 and model 2 is <0.05 , meaning that wt is a significant confounder. The p.value of the ANOVA test between model 2 and model 3 is <0.05 , meaning that hp is a significant confounder too but with a smaller effect on the am coefficient.

Diagnostic of residual



We examined the residuals for the model including weight and transmission.

- 1 The residual vs fitted values plot showed no systematic patterns and the points are apparently randomly spread around the 0, suggesting that linearity and homoscedasticity assumptions are met.
- 2 The normal QQ-plot indicates approximate normality of residuals, which only deviate on extremities.
- 3 The Scale-location plot confirmed roughly constant variance across fitted values.
- 4 Cook's distance did not reveal influential outliers.

Overall, diagnostic checks support the adequacy of the model.

Conclusions

Our analysis indicates that manual transmission cars are generally more fuel efficient than automatic cars. In unadjusted comparisons, manuals achieved on average ~7

MPG more than automatics. However, after adjusting for weight—a strong confounder—the estimated benefit of manual transmissions was reduced to about +2.9 MPG.

This suggests that while manual transmission itself is associated with improved fuel efficiency, part of the apparent advantage arises from the fact that manual cars in this dataset are lighter. Residual diagnostics supported the validity of the chosen model, and the results are consistent with the hypothesis that manual transmission provides a modest but real fuel economy benefit.